



**PES University, Bengaluru**  
(Established under Karnataka Act 16 of 2013)

**END SEMESTER ASSESSMENT (ESA) - DEC 2023**

**UE19CS353 - Object Oriented Analysis and Design with Java**

**Total Marks : 100.0**

1.a. Which are the three types of constructors that can be defined in Java?  
Give examples for each. (6.0 Marks)

1.b. What is inheritance, explain different types of inheritance in java with  
an example for each. (6.0 Marks)

1.c. Briefly describe the need for garbage collector in java. (4.0 Marks)

1.d. Write the output of the following java program

```
class A {  
    void show() {  
        System.out.println("show of class A");  
    }  
}  
class B extends A {  
    void show() {  
        System.out.println("show of class B");  
    }  
  
    void disp() {  
        System.out.println("show of class B disp");  
    }  
}  
class supertest1 {  
    public static void main(String args[]) {  
        A aobj = new A();  
        B bobj = new B();  
        A refa;  
        refa = aobj;  
        refa.show();  
        refa = bobj;  
        refa.show();  
        if (refa instanceof B) {  
            ((B) refa).disp();  
        }  
    }  
}
```

(4.0 Marks)

2.a. Elucidate qualified association and aggregation relationship w.r.t class diagram with an example for each (6.0 Marks)

2.b. Describe in a line or two the following elements of component diagram: Port, Dependency, Component, Interface (4.0 Marks)

2.c. Draw an use case diagram for e-commerce website (6.0 Marks)

2.d. What are Generic classes in java. Write an example program to demonstrate any one generic class in java. (4.0 Marks)

3.a. i) Match the following Design Principle names with their definition. 4M

|                        |                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1. Information Expert  | a. Combine related responsibilities into one manageable unit.                                                               |
| 2. High Cohesion       | b. Provide a well defined interface so that there will be no affect on other units                                          |
| 3. Indirection         | c. Assign responsibility to the class which has the required data to fulfill that responsibility                            |
| 4. Protected Variation | d. Introduce an intermediate unit to communicate between the other units, so that the other units are not directly coupled. |

ii) Differentiate between guard condition and change event in state modeling. 2M (6.0 Marks)

3.b. Explain in a line or two the following GRASP principles with an example.

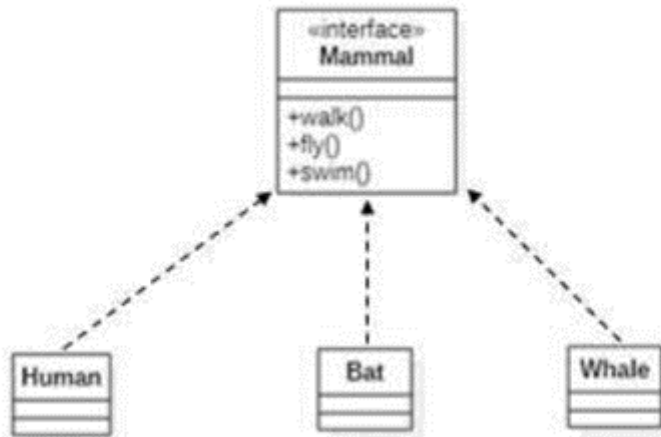
- i) Creator
- ii) Low Coupling
- iii) Controller
- iv) Polymorphism

(8.0 Marks)

3.c. Draw an activity diagram to perform withdrawal operation in an ATM machine. Specify all the activities and events along with swimlanes. (6.0 Marks)

4.a. What SOLID design principle is violated in the given design? How can this be modified to make it a better design? Draw class diagram and code snippet of your design.

(8.0 Marks)

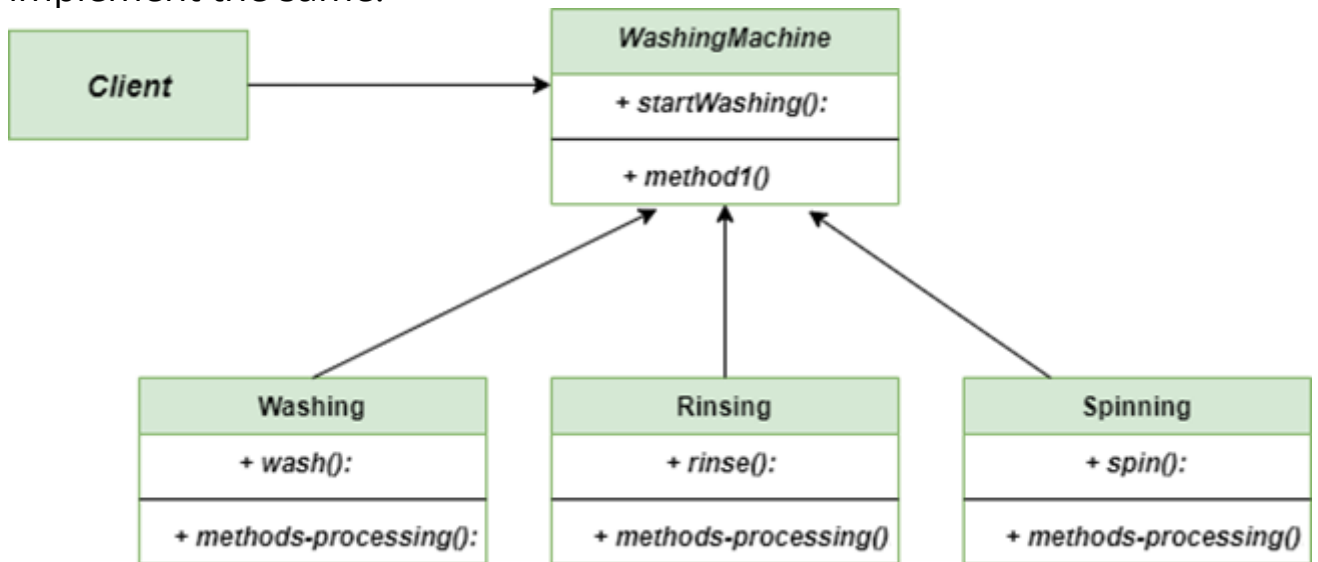


4.b. Describe the Singleton creational pattern with a suitable code snippet (6.0 Marks)

4.c. What are Design patterns? Which are the different types of design patterns, describe in a line or two? Specify pros and cons of design pattern. (2+2+2) (6.0 Marks)

5.a. Give the UML diagram of the Adapter pattern and briefly describe each component (7.0 Marks)

5.b. Which design pattern is depicted in the given diagram below? Describe the identified pattern in a line or two. Give the pseudo code to implement the same.



(8.0 Marks)

5.c. What is an Anti pattern? Briefly describe vendor lock-in anti pattern (5.0 Marks)